An interface that assembles itself.

A prompt-driven generative UI prototype, in the lineage of Google Disco / GenTabs.

Type a sentence. The app parses it deterministically into a layout schema and builds the screen on the fly — panels stagger in, a map springs up, a route draws itself, and shared elements **morph** between states instead of hard-cutting.

Disco's premise: the UI is the answer.

THE IDEA

Conventional search returns a list of documents. Disco/GenTabs returns an **assembled interface** — the response is a purpose-built screen for the thing you asked about, not a page of links.

The interface becomes a function of the query.

WHAT THIS PROTOTYPE PROVES

That the hard part is not the model. It is the **mapping layer** — turning language into a layout the renderer can trust — and the **motion** that makes a screen rebuilding itself feel intentional rather than broken.

Both are demonstrated here without an LLM in the loop.

No LLM is used. That is a deliberate scope decision, not a limitation — see slide 12.

Prompt text never touches a component.

One indirection layer decides everything.

Rather than branching on strings inside the UI, the prompt is compiled into a normalized, serializable Scene — a layout descriptor. The renderer only ever knows that schema. Every property of the system worth grading falls out of this one decision:

- → The parser is **pure and synchronous**, so it unit-tests without a browser.
- → The renderer is **registry-driven**, so new capabilities need no renderer edits.
- → Scenes are **data**, so state changes are tweens between two values — never a teardown.
- → The producer is **swappable**, so a real model can replace the parser later.

Prompt → Scene → screen.

Normalize casing, whitespace → **Classify** weighted keywords → **Extract** gazetteer match → **Reconcile** arity, conflicts → **Scene** layout schema → **Render** registry-driven

CLASSIFICATION IS LEGIBLE

Intent is a bag of weighted keywords, not a black box — tunable, inspectable, and the same mechanism that resolves conflicts. A prompt containing both "flight" and "hotel" resolves to the higher weighted score, deterministically.

```
kind: "flight", arity: 2,
keywords: { flight: 3, fly: 3,
            route: 2, trip: 1 }
```

ENTITIES ARE GEOGRAPHIC

A local gazetteer resolves places by longest-match-first, each carrying `lat/lng` and aliases. A shared `project()` maps geography into the stage's normalized 0–1 space using the **same equirectangular projection the map is drawn on** — so a projected marker lands on the matching landmass.

The layout descriptor.

```ts
// the contract between parsing and rendering
interface Scene {
  key:      string;   // stable render identity
  intent:   IntentKind;
  prompt:   string;
  headline: string;   // narrates the "generation"
  stage:    StageConfig;  // camera + markers + route
  detail:   DetailPanel | null;
  chips:    string[];
}
```

01 · STABLE PANEL IDENTITY

`DetailPanel.id` is always `"detail"`, whatever the variant. Same id ⇒ the renderer treats a flight panel and a hotel panel as **the same physical element** ⇒ FLIP morph, not unmount/remount.

02 · CAMERA AS DATA

The map never re-instantiates. Scenes only change `stage.camera` and `stage.markers/route`, so every visual change is a tween between two schema values.

Intent, not pixels.

Where the boundary gets interesting.

THE PROBLEM

A builder *cannot* compute the zoom that frames a route — the answer depends on the container's aspect ratio at render time, which the parser must not know about.

THE RESOLUTION

The descriptor states the **requirement**, not the value: `camera.fit`, a normalized rectangle that must stay in shot. `MapStage` resolves it against a `ResizeObserver`-measured container, treating the authored `zoom` as the tightest allowed framing.

New York → London keeps its authored `1.25`. San Francisco → Sydney pulls back until both cities are on screen, at any viewport. Single-focus scenes omit `fit` entirely.

This is the boundary working as designed: the parser stays pure and deterministic — and therefore unit-testable — while everything that needs to know about pixels stays in the renderer.

The Transition.

"Show my flight route from New York to London"

01

The centred hero **prompt bar contracts and repositions** to the top strip — one mounted element, FLIP-slid, never a swap.

02

The **map stage scales up** into the workspace on a spring camera move.

03

An **SVG route draws itself** — animated `pathLength` from the New York marker to the London marker.

04

The **detail rail slides in** and its fields enter staggered: flight number, duration, landing time.

The Morph.

…then "Show me my hotel details for London"

- → **Nothing unmounts.** No flash, no reflow, no reset.
- → The route **fades out**; the camera zooms from the transatlantic framing into the London region.
- → The detail rail **keeps its identity** — shared `layoutId="detail-panel"` — and FLIP-resizes, its flight stats cross-fading into a hotel check-in card inside the re-flowed container.

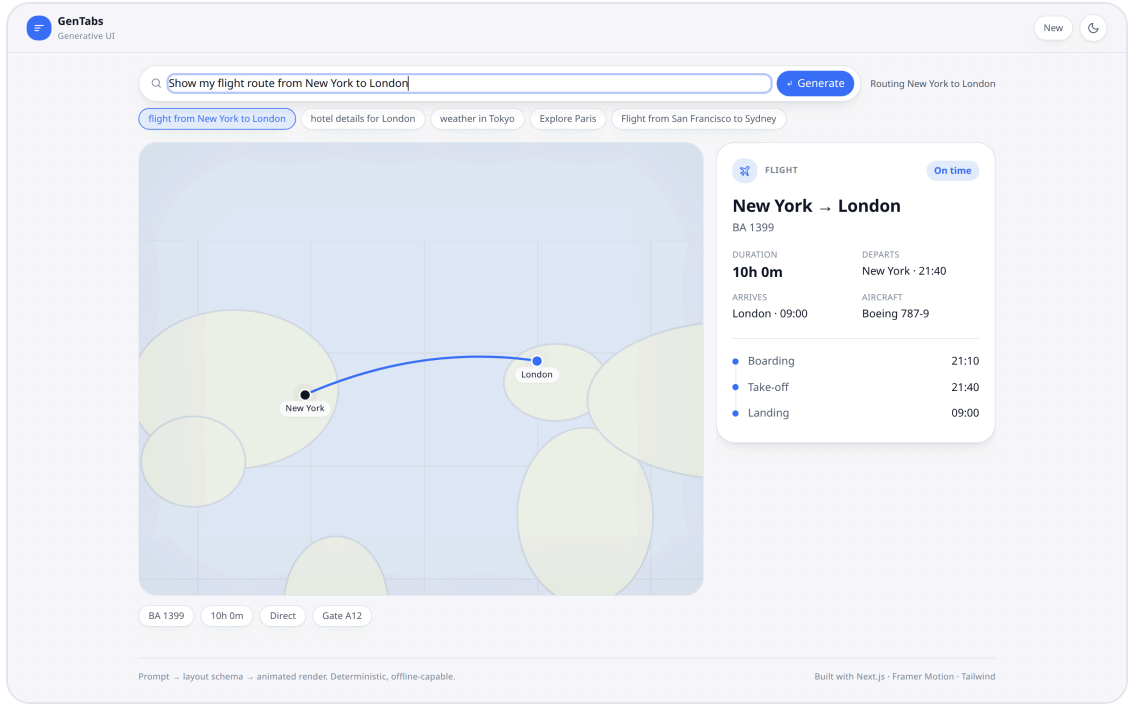
ENFORCED, NOT HOPED FOR

Continuity is the explicit goal, so the invariant that makes it possible is covered by a test rather than left to convention:

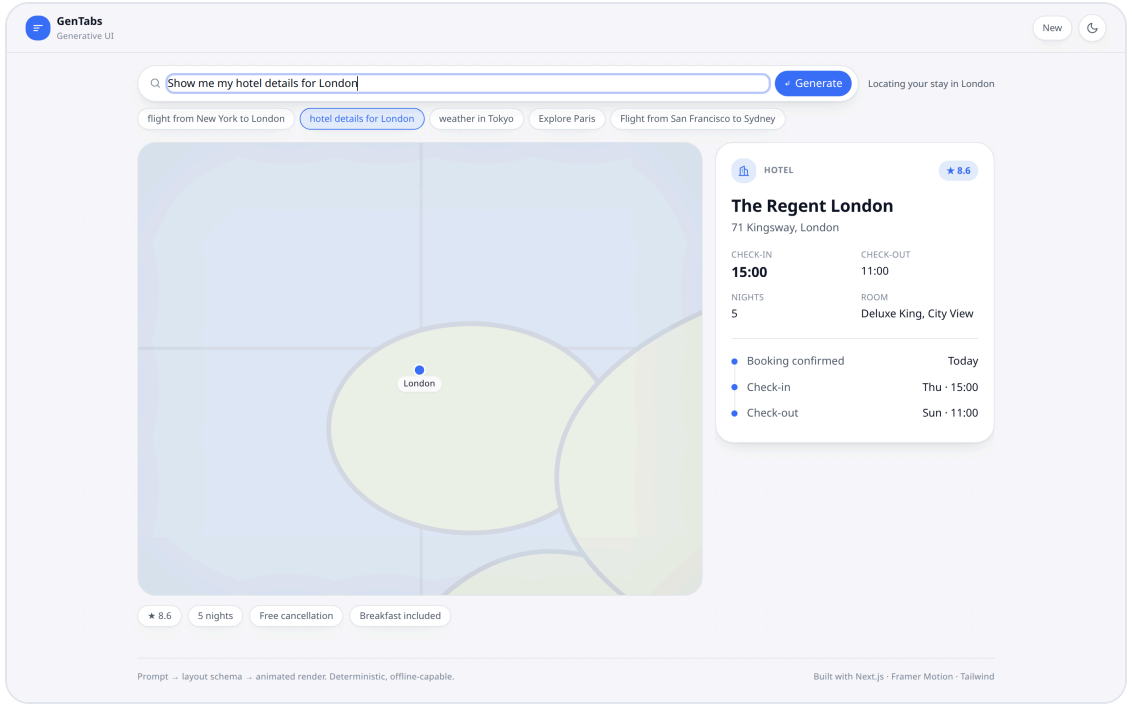
```
✓ panel id is stable across
  flight→hotel (enables morph)
```

If a future change breaks the shared identity, the morph silently degrades into a hard cut — the suite catches it first.

Two prompts, back to back.



The Transition — prompt bar docked, camera framed to the route, SVG path drawn, detail rail staggered in.



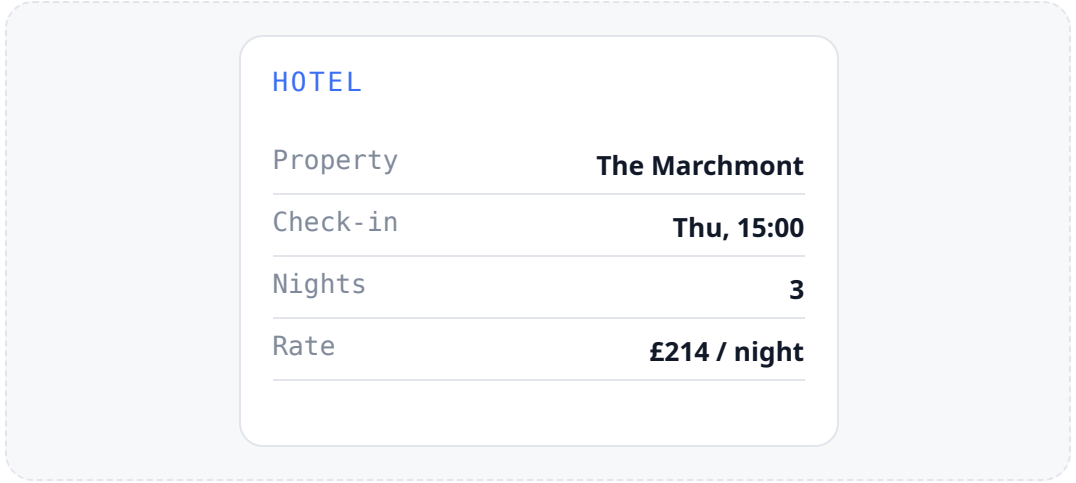
The Morph — same rail, same mount. Route retired, camera zoomed to London, contents cross-faded to the hotel card.

One physical language.

Framer Motion was chosen over GSAP or WAAPI for one decisive reason: **`layout` + `layoutId` give FLIP for free** — exactly the primitive the Morph demands. It also ships `useReducedMotion()`, keeping accessibility inside the motion system rather than bolted on.

TOKEN	USE	CHARACTER
`spring.press`	hover / tap	`520 / 30`
`spring.layout`	morphs, docking	`240 / 30`
`spring.entrance`	panels, stagger	`260 / 32`
`spring.camera`	map zoom / pan	`120 / 26`

Components import these instead of hand-tuning numbers. Only compositor-friendly properties animate — transform, opacity, pathLength.



A standalone illustration of the primitive — the container persists and resizes while its contents cross-fade.

Four states, deliberately staged.

idle
hero
→
parsing
420ms · "Understanding…"
→
generating
620ms · skeleton / shimmer
→
ready
staggered reveal

WHY STAGE IT AT ALL

Instant results would read as a lookup. The machine gives the fake generation a legible rhythm: skeletons occupy the **final geometry first**, so revealed content never shifts the layout, and the reveal reads as the container filling in.

Content then enters with `staggerChildren: 0.07`, blur-to-sharp.

WHY IT PRODUCES A MORPH FOR FREE

On a follow-up prompt the previous scene's panels **stay mounted**. The machine's next `generating → ready` cycle is therefore expressed as a morph rather than a reset — the continuity is structural, not special-cased.

Under `prefers-reduced-motion` the same timings collapse to 60ms / 90ms.

Design system & accessibility.

TOKENIZED, THEMEABLE

Semantic CSS variables — `--ink`, `--surface`, `--accent` — never raw hex in components, so the entire palette re-themes by swapping one variable block. A single modular type scale, 8pt spacing rhythm, tokenized elevation and radius.

Body ink targets ≥7:1 contrast; muted text ≥4.5:1 (WCAG AA/AAA).

ACCESSIBLE BY CONSTRUCTION

Semantic landmarks and `dl/dt/dd`; skip-to-content; visible tokenized focus rings. The map carries a descriptive `role="img"` label ("Map showing a route from New York to London") while decorative SVG is hidden.

The lifecycle is narrated through an `aria-live` region, so generation is perceivable without sight.

Reduced motion is honoured in two layers — a global CSS reset neutralizing shimmer and transitions, *and* `useReducedMotion()` in code, which swaps positional animation for plain opacity and shortens the machine. Responsive throughout: below `lg` the same panels reflow to a stacked column rather than re-mounting.

This deck is built from those same tokens — it inherits the app's palette, type stack and dark theme rather than restating them.

The parser is a placeholder by design.

Today `parsePrompt()` → **Scene** `layout schema` ← **Tomorrow** `LLM streaming JSON`

Because the renderer consumes a **serializable schema** rather than prose, the producer behind it is interchangeable. A model emitting Scene-shaped JSON drops into the same seam — **no renderer changes** — and everything already built keeps working.

WHAT THAT UNLOCKS

- → Open-vocabulary prompts, without a gazetteer ceiling.
- → **Progressive rendering** — panels appear as the JSON streams, which the staged machine is already shaped for.
- → Server-authored scenes, since `Scene` is plain data and crosses the wire unchanged.

WHAT STAYS HONEST

The schema is also the **safety boundary**. A model proposes a layout; it never ships executable UI. Unknown variants fall back rather than crash, and every scene remains inspectable, diffable and testable — the same properties that make the deterministic parser trustworthy today.

See it running.

15 / 15
parser tests passing

0
type errors

Static
every route prerenders

2
themes, both designed

google-disco.netlify.app
Live demo — try the two prompts back to back to see the morph.

github.com/swmd/google-disco
Source, README and the full design document.

docs/DESIGN.md
User flow, mapping architecture, motion strategy.

No API keys, no database, no server runtime — a CDN-served build.